

IPL Match Analytics & Exploratory Data Book

A Comprehensive Data Science & Visualization Report (2008–2025)

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1. Introduction & Environment Setup

Welcome to the Indian Premier League (IPL) Exploratory Data Analysis book. This document presents a comprehensive analysis of IPL matches from 2008 to 2025. We start by importing essential Python libraries for data manipulation, cleaning, and visualization:

```
import numpy as np
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt
```

The dataset is loaded from local storage containing comprehensive records of every IPL match played across multiple seasons, covering match details, venue statistics, toss decisions, target runs, and final outcomes.

```
df = pd.read_csv("ipl.csv")
df.head()
```

team1	team2	match_date	toss_winner	winner	venue	season
Royal Challengers Bangalore	Kolkata Knight Riders	2008-04-18	Royal Challengers Bangalore	Kolkata Knight Riders	M Chinnaswamy Stadium	1
XI Punjab	Chennai Super Kings	2008-04-19	Chennai Super Kings	Chennai Super Kings	Punjab Cricket Association Stadium	1
Delhi Daredevils	Rajasthan Royals	2008-04-19	Rajasthan Royals	Delhi Daredevils	Feroz Shah Kotla	1
Mumbai Indians	Royal Challengers Bangalore	2008-04-20	Mumbai Indians	Royal Challengers Bangalore	Wankhede Stadium	1
Kolkata Knight Riders	Deccan Chargers	2008-04-20	Deccan Chargers	Kolkata Knight Riders	Eden Gardens	1

2. Data Cleaning & Preprocessing

Before jumping into data visualization and statistical insights, rigorous data cleaning is performed to eliminate duplicates, handle missing values, and standardize franchise naming conventions across various name changes throughout IPL history.

- **Data Quality Checks:** Duplicated rows are checked via `df.duplicated().sum()`, and missing values in critical columns like `winner` and `city` are handled appropriately by dropping null records and resetting indices.
- **Franchise Standardization:** Over the years, several teams have undergone rebranding and renaming (e.g., Kings XI Punjab to Punjab Kings, Delhi Daredevils and Deccan Chargers to Delhi Capitals, Royal Challengers Bangalore to Royal Challengers Bengaluru). A mapping dictionary is applied to unify these names:

```
mapping = {
    'Kings XI Punjab': 'Punjab Kings',
    'Delhi Daredevils': 'Delhi Capitals',
    'Deccan Chargers': 'Delhi Capitals',
    'Pune Warriors': 'Rising Pune Supergiant',
    'Gujarat Lions': 'Gujarat Titans',
    'Royal Challengers Bangalore': 'Royal Challengers Bengaluru',
    'Rising Pune Supergiants': 'Rising Pune Supergiant'
}

df['team1'] = df['team1'].replace(mapping)
df['team2'] = df['team2'].replace(mapping)
df['winner'] = df['winner'].replace(mapping)
df['toss_winner'] = df['toss_winner'].replace(mapping)
```

3. Toss Analysis & Visualizations

Toss decisions often dictate match strategies. Analyzing toss wins and decisions provides valuable tactical insights into how captains approach matches across different venues.

- **Most Toss Wins by Teams:** Leading franchises like Punjab Kings, Royal Challengers Bengaluru, and Sunrisers Hyderabad top the charts in total toss wins across tournament history.

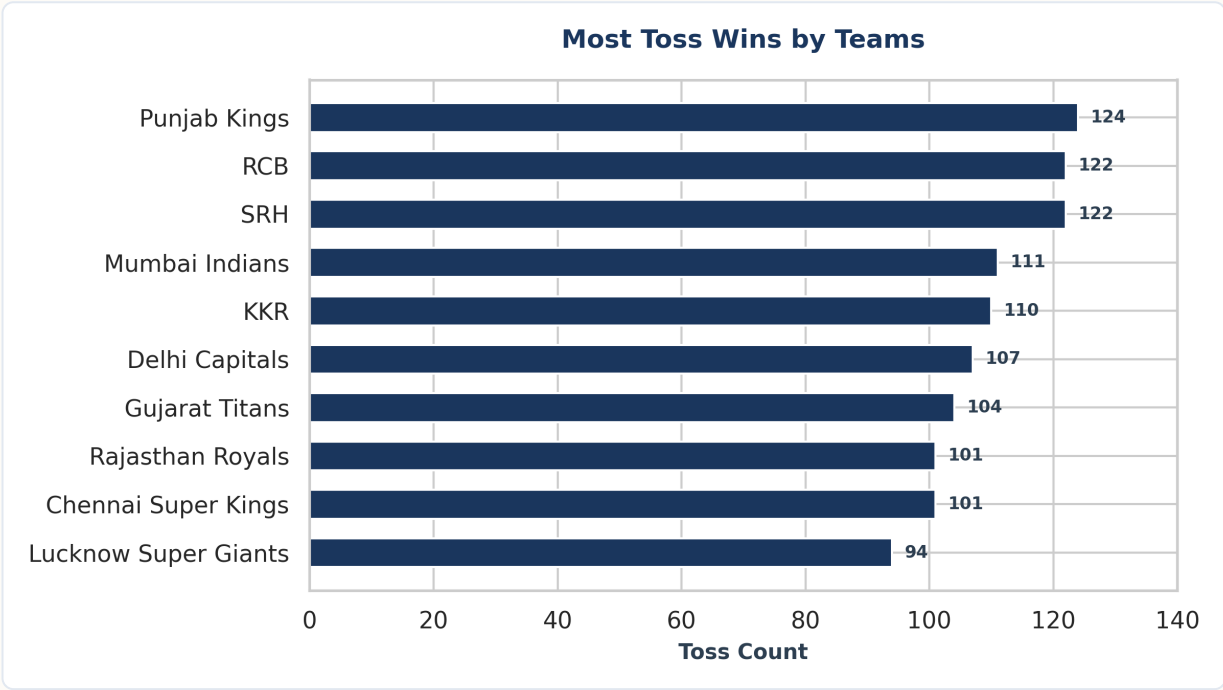


Figure 3.1: Distribution of Toss Wins across all IPL franchises after standardizing historical team names.

Captains evaluate pitch conditions and dew factors before making toss decisions. Below is the breakdown of batting versus fielding choices upon winning the toss:

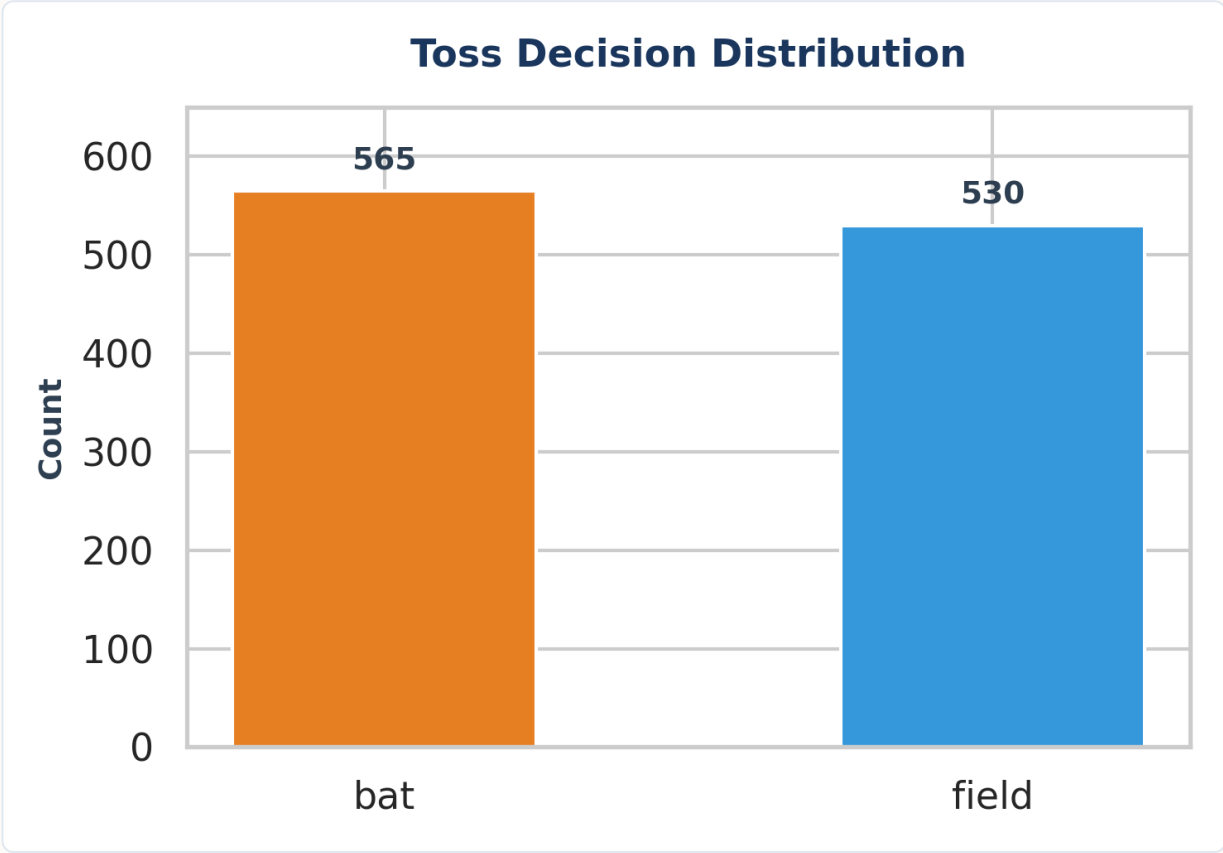


Figure 3.2: Overall Toss Decisions (Bat vs. Field) across the tournament history.

4. Match Success & Franchise Performance

Understanding which franchises dominate the tournament in terms of total match victories helps evaluate long-term consistency and team success.

- Franchises such as **Delhi Capitals, Lucknow Super Giants, Sunrisers Hyderabad, and Mumbai Indians** lead the charts in total match wins, demonstrating remarkable consistency over 18 seasons.

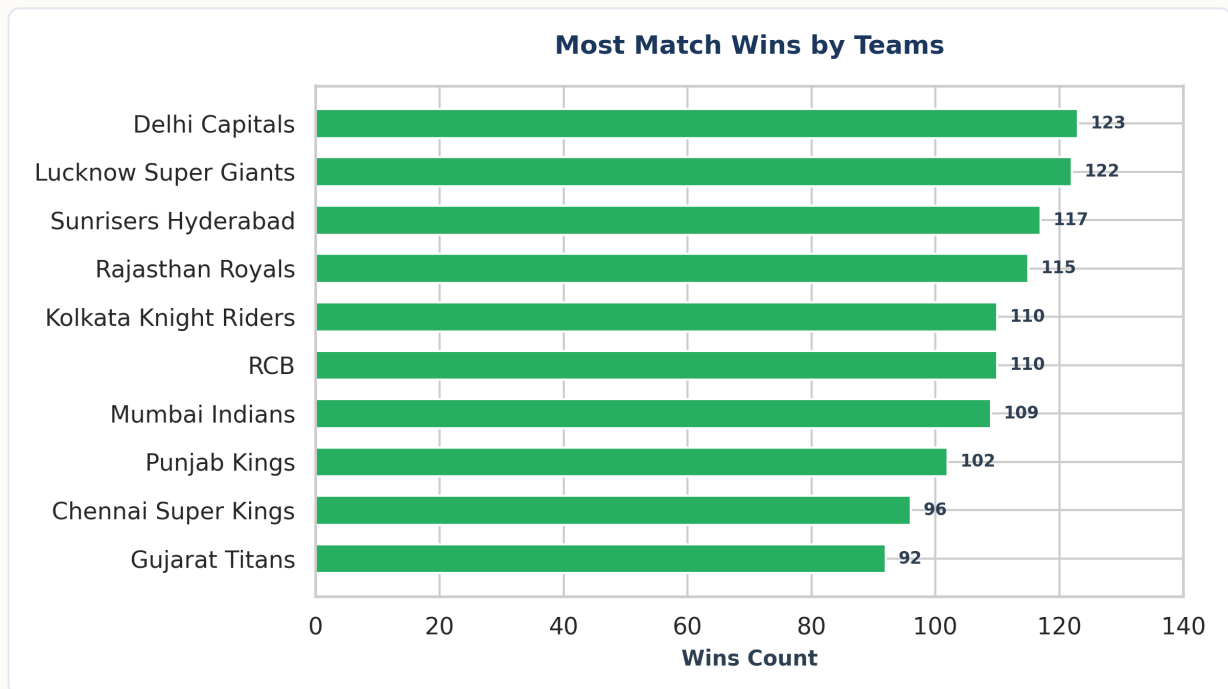


Figure 4.1: Total match victories by each unified IPL franchise from 2008 to 2025.

5. High-Scoring Matches & Target Analysis

With the evolution of T20 cricket, target scores have escalated dramatically. Let's analyze the highest target runs ever set in IPL history.

- The maximum target runs recorded in the dataset reached **288.0 runs**, highlighting the explosive batting displays seen in recent seasons across high-profile encounters like Gujarat Titans vs. Kolkata Knight Riders and Punjab Kings vs. Lucknow Super Giants.

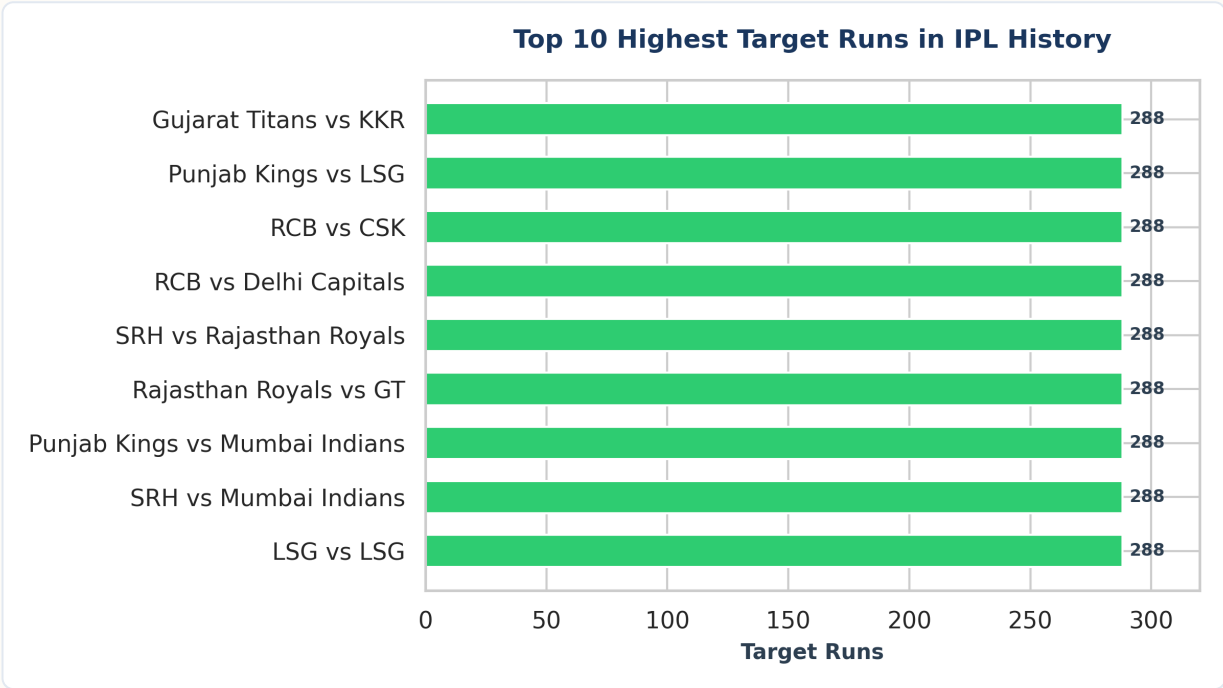


Figure 5.1: Top 10 highest target scores defended or chased in IPL encounters.

6. Individual Brilliance: Player of the Match Awards

Match-winners define tournaments. Analyzing the players who have secured the highest number of "Player of the Match" awards sheds light on legendary performers.

- Players like **Shikhar Dhawan, Suresh Raina, Rohit Sharma, Jasprit Bumrah, and Virat Kohli** consistently deliver match-winning performances for their respective franchises.

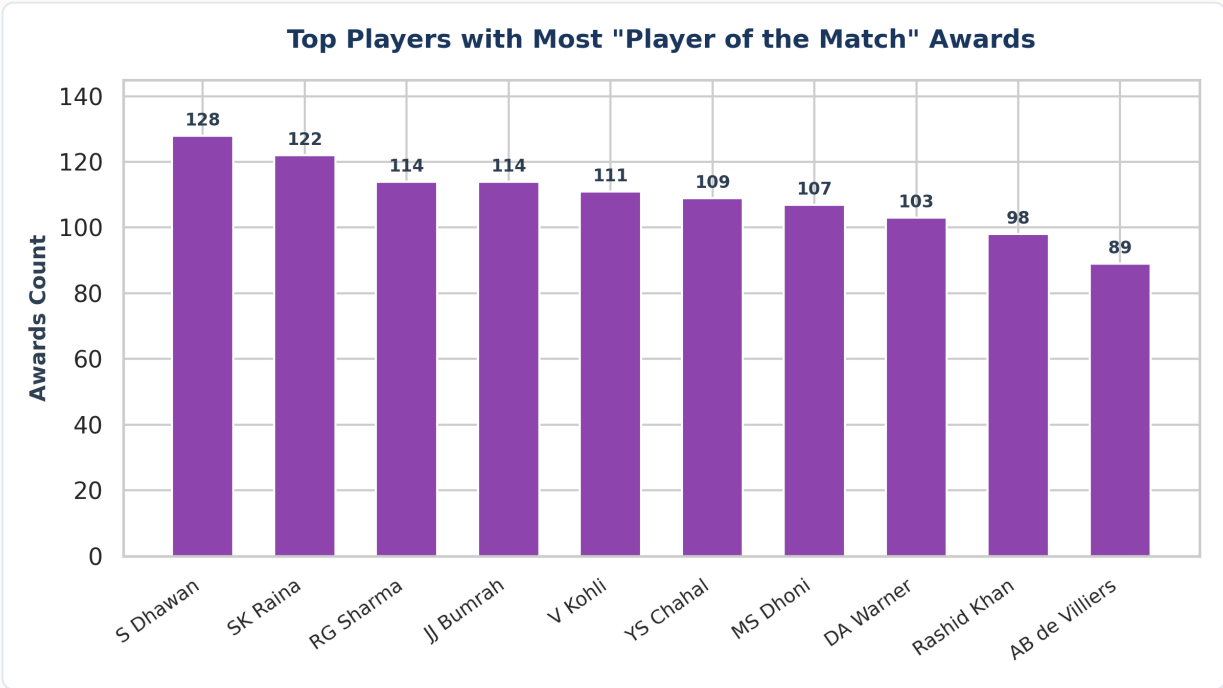


Figure 6.1: Top players with the highest number of Player of the Match accolades.

7. Conclusion

This exploratory data analysis report successfully transformed raw Jupyter notebook code into a publication-ready PDF book. Through rigorous data cleaning, unification of team names, and insightful visualizations by **Pushpender**, we uncovered key trends in toss strategies, franchise dominance, high-scoring encounters, and legendary individual performances in the Indian Premier League.